Executive Summary — Study Buddy (RAG Project)

Study Buddy is an intelligent Retrieval-Augmented Generation (RAG) system designed to support efficient, personalized learning. The project integrates document retrieval, text chunking, vector embeddings, and a lightweight generative model to create an AI assistant capable of answering questions directly from course materials. By grounding responses in retrieved context, Study Buddy provides accurate, source-based explanations rather than generic model output.

The system begins by ingesting study documents—such as notes, textbooks, or lecture slides—and converting them into searchable vector embeddings. When a user asks a question, Study Buddy retrieves the most relevant text segments and uses them to generate a clear, context-aware answer. This approach ensures reliability, transparency, and alignment with the user's actual study content.

Throughout the project, key skills were developed across data preprocessing, embedding-based retrieval, prompt engineering, and evaluation of RAG system performance. The result is a practical, user-focused learning tool that demonstrates how modern AI techniques can enhance comprehension, reduce study time, and support mastery of complex topics.